

Cosmic Expansion and Gravity from Recursive Quantum Dynamics

Abstract

We present a self-contained theoretical framework in which quantum mechanics, inertia, gravity, and cosmological expansion emerge from a single underlying principle: recursive dynamics with finite memory. Time is not assumed a priori; instead, it arises as a continuum limit of discrete recursion. We show that the Schrödinger equation appears as a parabolic limit of recursion, while relativistic dynamics arise from hyperbolic recursion. Mass is identified with memory persistence, leading to a natural equivalence between inertial and gravitational mass. Cosmological expansion is derived as a global recursion drift, yielding Friedmann equations and, under locality and covariance, the Einstein field equations. The cosmological constant Λ emerges as surplus recursion not bound into localized memory (structure), resolving the vacuum energy problem. The framework makes concrete, falsifiable predictions for late-time structure growth, offering potential explanations for current tensions in H_0 , σ_8 , and growth-rate measurements.

1. Introduction

Modern physics rests on three pillars—quantum mechanics (QM), general relativity (GR), and cosmology—whose conceptual foundations remain disjoint. Quantum theory presupposes a global time parameter, while gravity renders time dynamical. Cosmology introduces a cosmological constant Λ whose physical origin remains obscure. These tensions suggest that time, mass, and gravity may not be fundamental but emergent.

In this work we propose a unifying framework based on *recursive dynamics with finite memory*. The guiding idea is simple: physical reality is described by states evolving through discrete recursion steps. Stable structures emerge as attractors sustained by memory. Causation, time, mass, and geometry arise only in appropriate limits. We demonstrate that this single mechanism reproduces known physics and yields new, testable cosmological predictions.

2. Recursive Dynamics and Emergent Time

2.1 Fundamental recursion

Let $\Psi(x, n)$ denote a state defined on configuration space x and recursion index n . Evolution is given by

$$\Psi_{n+1} = \mathcal{R}[\Psi_n],$$

where $\mathcal{R}$ is a recursion operator. No notion of physical time is assumed.

We consider weakly coupled recursion,

$$\mathcal{R} = I + \epsilon \mathcal{L} + O(\epsilon^2), \quad \epsilon \ll 1,$$

with $\mathcal{L}$ a linear operator encoding local propagation of memory.

2.2 Continuum limit

Define an effective time parameter

$$t := n\epsilon.$$

Then

$$\partial_t \Psi = \mathcal{L} \Psi.$$

Time emerges as a bookkeeping parameter for accumulated recursion.

3. Schrödinger Equation as a Limit

3.1 Stability and unitarity

To ensure stability of memory (no exponential blow-up or decay), $\mathcal{L}$ must be anti-Hermitian:

$$\mathcal{L}^\dagger = -\mathcal{L}.$$

Thus

$$\mathcal{L} = -\frac{i}{\hbar_{\text{eff}}} \hat{H},$$

leading to

$$i\hbar_{\text{eff}} \partial_t \Psi = \hat{H} \Psi.$$

3.2 Spatial locality

The lowest-order, translation-invariant choice is

$$\hat{H} = -\frac{\hbar_{\text{eff}}^2}{2\mu} \nabla^2 + V(x),$$

where μ characterizes resistance to recursive dispersion. This yields the Schrödinger equation without postulating it.

4. Mass as Memory Persistence

4.1 Definition

Under free evolution, the spatial variance of Ψ grows diffusively. We define the memory parameter μ via the diffusion constant

$$D = \frac{\hbar_{\text{eff}}}{\mu}.$$

Large μ corresponds to slow dispersion and strong localization.

Definition: Mass is the persistence of memory under recursion,

$$m \equiv \mu.$$

4.2 Inertial mass

The parameter μ controls the response of Ψ to forces, reproducing inertial mass in the Schrödinger equation.

5. Equivalence of Inertial and Gravitational Mass

5.1 Gravity as recursion distortion

Allow the recursion step size to vary spatially,

$$\epsilon \rightarrow \epsilon(x).$$

This induces an effective metric perturbation

$$\Phi(x) = \frac{\delta\epsilon(x)}{\epsilon}.$$

5.2 Memory as gravitational source

Define memory density

$$\rho_\mu = \mu |\Psi|^2.$$

Variations of recursion density sourced by ρ_μ yield

$$\nabla^2 \Phi \propto \rho_\mu,$$

recovering Poisson's equation. The same parameter μ governs inertia and gravitational sourcing:

$$m_{\text{inertial}} = m_{\text{gravitational}}.$$

The equivalence principle emerges as a consequence of uniform recursion rescaling.

6. Relativistic Extension via Hyperbolic Recursion

6.1 Second-order recursion

To enforce finite propagation speed, we consider

$$\Psi_{n+1} - 2\Psi_n + \Psi_{n-1} = \epsilon^2 \mathcal{K} \Psi_n.$$

In the continuum limit,

$$\partial_t^2 \Psi = \mathcal{K} \Psi.$$

6.2 Klein–Gordon equation

Lorentz invariance fixes

$$\mathcal{K} = c^2 \nabla^2 - \omega_0^2,$$

leading to

$$(\partial_t^2 - c^2 \nabla^2 + \omega_0^2) \Psi = 0.$$

With $\omega_0 = mc^2/\hbar$, rest mass is identified with intrinsic recursion oscillation frequency.

7. Cosmological Expansion as Global Recursion Drift

7.1 Global recursion rate

Allow the recursion rate to drift globally,

$$\epsilon \rightarrow \epsilon(t).$$

Define the scale factor by

$$\frac{1}{a} \frac{da}{dt} = H(t) = \gamma \epsilon(t).$$

Cosmic expansion is thus identified with global recursion drift.

7.2 Friedmann equations

Memory binding into structure slows recursion, while surplus recursion drives expansion. This balance yields

$$H^2 = \frac{8\pi G}{3} \rho + \frac{\Lambda}{3},$$

$$\dot{H} = -4\pi G(\rho + p).$$

Here Λ is surplus recursion not bound into memory, not vacuum energy.

8. Emergence of Einstein Equations

The Friedmann equations constitute the homogeneous sector of GR. Imposing locality, covariance, and conservation of memory ($\nabla^\mu T_{\mu\nu} = 0$), their unique covariant extension is

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}.$$

General relativity emerges as the hydrodynamic limit of recursive dynamics.

9. Late-Time Predictions and Observational Tests

9.1 Growth of structure

Memory leakage reduces effective clustering strength, modifying the growth equation

$$\ddot{\delta} + 2H\dot{\delta} - 4\pi G_{\text{eff}}(t)\rho\delta = 0,$$

with $G_{\text{eff}} < G$ at late times.

Prediction: Growth index

$$\gamma \approx 0.56\text{--}0.60 \quad (z < 1),$$

measurable by DESI, Euclid, and LSST.

9.2 Additional signatures

- Suppressed $\sigma_8 \approx 0.76\text{--}0.79$
- Mild redshift evolution of $w(z) > -1$
- Enhanced late-time ISW effect

10. Discussion and Outlook

We have presented a unified, self-contained framework in which QM, GR, and cosmology arise from recursive dynamics with finite memory. The cosmological constant is reinterpreted as surplus recursion, avoiding the vacuum energy catastrophe and yielding falsifiable predictions.

Future work will derive the full energy-momentum tensor, explore black hole entropy as memory saturation, and compute cosmological perturbations in detail.

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